Museum Monitoring: an environmental monitoring dataspace using The Things Network, Solid, and LDES

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Abstract

Museums increasingly deploy environmental monitoring systems (e.g., data loggers and sensor networks) to support preservation of cultural heritage objects. However, monitoring data is often captured and accessed through proprietary vendor software, which makes reuse, integration, and controlled cross-institution sharing difficult—particularly in object loan scenarios. This paper reports on MuMo (Museum Monitoring), a three-year applied research project that explored how dataspace principles can be applied to environmental monitoring in real museum settings. Rather than replacing existing systems, MuMo extends a legacy monitoring dashboard with semantic data modeling, Linked Data Event Streams, and Solid-compliant data access management. We present the system design, where environmental monitoring data is semantically described and published with access controlled fragments as a stream. The semantic links between these fragments allow clients to prune irrelevant branches while providers can restrict access at natural boundaries (e.g., per location or sensor). Further, we describe how MuMo is used in practice through a set of in-use scenarios, where the system enables (1) decentralized data publication for longitudinal analysis of environmental conditions, (2) selective cross-institutional data sharing during object loans, and (3) client-side data integration and aggregation across independent deployments. In practice, this dataspace-oriented deployment reveals design trade-offs while preserving institutional autonomy: museums control their data and authorization policies end-to-end, supporting trusted data sharing across organizational boundaries. The deployed system and its insights are thus relevant to a broad range of (Digital Humanities) projects that involve long-running data integration under distributed governance.

Keywords

Digital Humanities, Dataspaces, Solid, LDES, Museum Monitoring

1. Introduction & Motivation

In cultural heritage institutions, environmental parameters such as temperature, relative humidity, and light exposure directly influence the long-term preservation of collection objects [13]. Museums therefore invest in digital infrastructures for continuous environmental monitoring and traceable reporting [13, 16].

A wide range of sensing technologies is already used in practice, including data loggers and long-running wireless sensor network deployments in museum buildings [17]. Yet these systems typically remain tightly coupled to proprietary (and often legacy) dashboards or local infrastructures [1]. As a consequence, monitoring data is frequently siloed: it is difficult to combine measurements across installations, to align monitoring data with other institutional sources, or to selectively share a well-scoped subset of measurements with external partners under enforceable access control. This fragmentation reduces the analytical and documentary value of monitoring data and makes cross-organizational reuse unnecessarily costly.

This paper reports on MuMo (Museum Monitoring)[1], a three-year applied research project[2], funded by the Flemish Government, that investigates how museum environmental monitoring can

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[1] https://faro.be/project/museum-monitoring-tool

[2] https://www.vlaanderen.be/cjm/nl/cultuur/cultureel-erfgoed/subsidies-cultureel-erfgoed/projectsubsidies-cultureel-erfgoed/internationale-landelijke-en-bovenlokale-cultureel-erfgoedprojecten

be made more reusable, interoperable, and selectively shareable across organizational boundaries—without forcing museums to replace their day-to-day operational tooling. The project included both hardware/firmware and a software dataspace layer; this paper focuses on the software and dataspace aspects.

MuMo is motivated by monitoring needs that can be summarized as three usage modes:

- Operational oversight Staff need near-continuous insight into readings, time-series views, and alerts when conditions leave acceptable ranges.
- Long-term documentation Museums need access to historical exposure conditions over extended periods, so they can reconstruct how an object's environment evolved across locations and throughout its lifecycle.
- Selective collaboration and sharing When multiple parties are involved—most notably during loans—access must be manageable and bounded [11]. Practically, this means permissions that can be limited by (i) a subset of sensors/rooms, (ii) a defined time window, and (iii) the involved organizations.

The loan scenario is a key stress test. Loan agreements frequently include environmental constraints[3], and the lending institution needs trustworthy insight into the conditions experienced by an object while it is hosted elsewhere. At the same time, the borrowing institution is typically unable—and often unwilling—to expose its full internal monitoring landscape. In practice, this leads to ad-hoc exports and manual reporting, introducing delays, duplicate effort, and reduced transparency [9, 8].

This paper makes three contributions: (1) an in-use account of deploying dataspace principles for environmental monitoring in museums, (2) a concrete architectural integration of semantic modeling, event-based publication, and decentralized governance with legacy systems, and (3) empirically grounded lessons on aligning access control and data sharing mechanisms with institutional practice.

After describing the project context in Section 2, we present MuMo's architecture and approach in Section 3, and illustrate its use in practice through in-use scenarios in Section 4, before discussing lessons learned and implications for the Digital Humanities community.

2. Background and Design Constraints

To avoid ambiguity, we use MuMo v1 for the earlier prototype and MuMo v2 for the work presented in this paper. Unless stated otherwise, "MuMo" refers to MuMo v2.

MuMo v1 originated from the observation that environmental monitoring data in museums is difficult to access and reuse outside vendor dashboards. To assess what could be achieved with modest resources, the Fashion Museum Antwerp built an initial prototype to collect measurements and visualize them in a lightweight dashboard. This monitoring setup centered on a Raspberry Pi connected to a PHP dashboard. Based on the prototype's demonstrated potential, the museum applied for funding to extend the work.

2.1. Starting point: MuMo v1

MuMo v2 starts from MuMo v1, which already supported core operational monitoring: ingestion and persistence of sensor readings, time-series visualization, alerting, and basic export functionality. MuMo v1's dashboard also supported user and group management aligned with museum workflows, enabling staff to manage access within an institution in terms of responsibilities and spaces. In practice, these groups reflect a nested location hierarchy: from the museum or site level down to individual rooms and storage areas, and—when needed—even to fine-grained containers such as cabinets, shelves, or specific boxes.

[3] https://collectionstrust.org.uk/wp-content/uploads/2017/11/Loans-out-lending-objects.pdf, p11

At the same time, MuMo v1 made visible the limitations that motivated MuMo v2's focus: data remained largely bound to a single dashboard instance, exports were the primary sharing mechanism, and interoperability and cross-institution governance were not first-class concerns.

2.2. Scope: extending rather than replacing existing systems

MuMo v2 is explicitly not a "rip-and-replace" effort. Museums already depend on established dashboards and workflows that are difficult to change in day-to-day practice. Consequently, MuMo v2 keeps a dashboard-centric workflow as the primary operational interface for staff—where sensors are configured, readings are inspected, and alerts are handled—while adding a complementary data layer aimed at long-term reuse and cross-institution sharing.

This design stance reflects practical constraints in museum IT: successful changes must remain compatible with institutional autonomy, limited technical staffing, and long-running deployments. MuMo v2 therefore preserves the operational loop ("measure → view → react") while enabling monitoring data to be reused and selectively shared beyond a single dashboard when needed.

2.3. What MuMo v2 adds: new hardware and a dataspace-oriented layer

Building on this baseline, MuMo v2 advances both hardware and software while keeping dashboard-centric workflows intact. On the hardware side, the project develops custom ultra-low-power sensors designed for long battery life and integrates LoRaWAN-based transmission[4] to support robust deployments. Measurements captured by the MuMo v2 monitoring devices were transmitted to off-the-shelf gateways and routed through The Things Network, where they were then ingested into an existing (legacy) monitoring dashboard. This dashboard became the primary operational interface for museum staff and therefore strongly shaped how data could be accessed, interpreted, and shared. On the software side, MuMo v2 introduces the capabilities needed for reuse and selective sharing: (i) semantic representation of monitoring data, (ii) incremental/event-based publication, and (iii) an authorization approach that can operate across institutional boundaries while remaining administratively feasible for museum staff.

2.4. Summary: context as a design constraint

In MuMo v2, the central context is the combination of (1) long-running, append-only monitoring data, (2) operational tooling that cannot be replaced, and (3) collaboration scenarios that require selective, revocable access across organizational boundaries. MuMo v2 is therefore shaped by a pragmatic objective: introduce interoperability and governance-aware sharing while preserving institutional autonomy and established workflows.

3. System and Approach Overview

Chapter 3 explains how MuMo turns those three needs from the introduction into a concrete system: it keeps the familiar day-to-day monitoring workflow intact, while making the resulting data durable enough to interpret years later and scoped enough to share only what a partner should see. The core idea is to treat monitoring not as a dashboard export, but as a governed dataset that can be reused across time and across institutions—especially in loan situations where transparency is necessary, but full internal exposure is not.

We begin by introducing the data representation MuMo relies on and the way monitoring observations are captured and published over time. Next, we describe how access and sharing are controlled so that institutions can expose selected views without relinquishing ownership.

[4]https://www.thethingsnetwork.org/docs/lorawan/architecture/

Finally, we walk through the concrete components and their interaction to show how these ideas come together in the running system.

3.1. Data model and semantic representation

Environmental monitoring data is only useful if it remains interpretable over long periods, including across sensor redeployments and organizational boundaries. MuMo therefore transforms the information already managed in the legacy dashboard into a semantic representation: both environmental observations (e.g., temperature, humidity, light exposure) and sensor configuration metadata (e.g. group/location).

3.1.1. Modeling sensors and observations

MuMo builds on established models for describing sensors and observations, most notably the W3C Semantic Sensor Network ontology (SSN/SOSA) [5, 10]. SSN/SOSA is particularly suitable because it separates (i) the sensor, (ii) the observation, and (iii) the feature of interest being observed. This allows MuMo to represent sensor redeployment and contextual change (e.g., a sensor moved to a different space) without rewriting historical observations that were produced under earlier conditions.

MuMo instantiates these concepts using the Flemish OSLO (Open Standaarden voor Linkende Organisaties) vocabularies [2], maintained under Digitaal Vlaanderen, which profile and reuse international semantic standards to support cross-organizational data exchange.

3.1.2. Representing changing context as configuration events

Monitoring data exhibits two different dynamics:

- Observations evolve continuously and are typically appended over time.
- Configuration context evolves discretely when sensors are moved or reconfigured.

MuMo makes this distinction explicit by publishing configuration metadata as a versioned event stream, allowing consumers to reconstruct which context applied when an observation was produced.

Group membership (or location) are encoded explicitly in this configuration stream as versioned entities. This is not merely descriptive metadata: it enables downstream systems to interpret observations correctly and consistently apply sharing rules that are expressed in terms museum staff already use (groups representing rooms, objects, loan packages, etc.).

3.2. Publication layer: Linked Data Event Streams

Museum monitoring datasets are long-running and effectively append-only: new measurements keep arriving, growing the stream. That “ever-growing log” is exactly what Linked Data Event Streams (LDES) are designed for—publishing data incrementally in fragments that consumers can efficiently synchronise over time. This approach is also actively promoted in Flanders (Digitaal Vlaanderen) for interoperable data sharing. MuMo therefore publishes monitoring data using LDES [6, 23].

LDES supports interoperable publication of evolving datasets in a way that enables replication and synchronization across organizational boundaries without requiring a centralized query service.

An LDES publication is typically organized as a fragment tree: events are partitioned into smaller resources (fragments), and each fragment exposes typed links to other fragments through machine-interpretable relations in the Linked Data representation. Listing 1 illustrates this with a first-level partitioning by sensor: the entrypoint fragment links, via `tree:EqualToRelation`

```turtle
@base <https://mumo.faro.be/data/> .
@prefix tree: <https://w3id.org/tree#> .
@prefix ldes: <https://w3id.org/ldes#> .
@prefix sosa: <http://www.w3.org/ns/sosa/>.

# LDES entrypoint
<stream>
  a ldes:EventStream ;
  tree:view <by-sensor> .

# <by-sensor> fragment
<by-sensor>
  a tree:Node ;
  tree:relation [
    a tree:EqualToRelation ;
    tree:path sosa:madeBySensor ;
    tree:value <https://mumo.faro.be/sensors/sensor-1> ;
    tree:node <by-sensor/sensor-1> ;
  ] ;
  tree:relation [
    a tree:EqualToRelation ;
    tree:path sosa:madeBySensor;
    tree:value <https://mumo.faro.be/sensors/sensor-2> ;
    tree:node <by-sensor/sensor-2> ;
  ] .
```

Listing 1: LDES fragmentation overview example, first fragmenting on location, then on group and lastly on time.

on `sosa:madeBySensor`, to a dedicated fragment for each sensor. In practice, such partitionings can be composed: once a client has navigated to the fragment for the relevant sensor, that fragment can in turn act as the root of a second tree (e.g., partitioned by time windows such as month/day), enabling clients to narrow down first by source and then by interval without scanning unrelated observations.

This supports three key consumption modes for monitoring data:

- retrieving historical data incrementally,
- staying synchronized with newly produced events, and
- avoiding repeated reliance on centralized query endpoints.

3.2.1. Fragmentation as a design lever

A central design choice in LDES is the fragmentation strategy: how events are grouped into fragments. MuMo uses fragmentation not only as a performance mechanism, but as a way to reflect how museums work with monitoring data and how sharing is scoped in practice.

In MuMo, observations are fragmented along the operational dimensions that conservator and technicians naturally use: group, sensor, and time. Concretely, observations are published under a hierarchy of the form:

```
group-x / sensor-x / year / month / day
```

This keeps fragments small and stable over time (e.g., daily fragments do not grow indefinitely), while retaining a navigable structure that allows clients to focus on the relevant group, sensors,

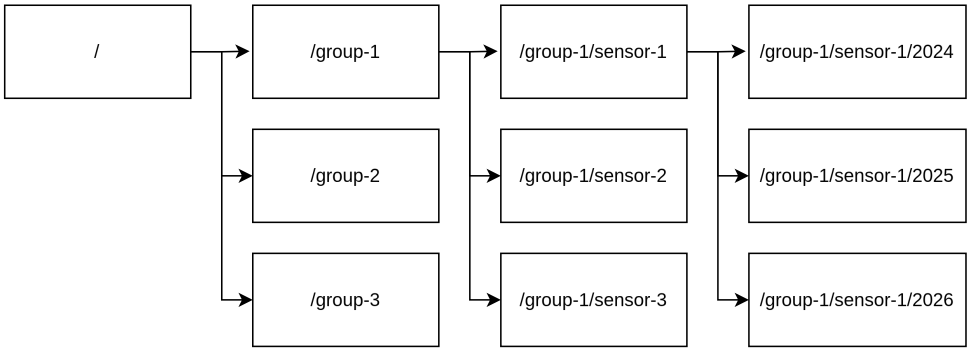

Figure 1: LDES fragmentation overview example, first fragmenting on location, then on group and lastly on time.

and time window. Note that the group-level fragmentation is flattened: the fragment tree does not encode the group hierarchy itself. Instead, parent–child relationships between groups are provided separately in the group metadata, which is sufficient for navigation.

Figure 1 illustrates the fragmentation strategy chosen for MuMo as a concrete example; we refer back to this tree throughout the remainder of the paper.

This publication model proved particularly suitable in cross-institutional settings because it allows consumers to process only the data they are authorized to access, without requiring providers to expose tailored query endpoints.

Prior work also demonstrates that LDES event sources can be stored and consumed within the Solid ecosystem, supporting the compatibility of event-stream publication with Solid-based access control. [19]

3.3. Governance layer: Solid-based decentralized sharing

Cross-institution collaboration (especially loans) requires selective, revocable access across organizational boundaries, while museums remain unwilling to relinquish operational control over their monitoring infrastructures.

MuMo addresses this by combining decentralized publication with Solid-based identity and authorization. Solid provides a framework in which storage, identity, and access control are decoupled from specific applications, enabling the same published resources to be reused by multiple clients without funneling all access through a centralized service [3, 15, 24].

3.3.1. Institutional endpoints and delegated access

In MuMo, Solid Pods act as institutional data endpoints rather than user-centric storage. The practical effect is that a museum can publish monitoring resources as long-lived datasets under its governance, while granting external parties access only to the parts required for a collaboration (e.g., a loan group) and revoking that access afterward.

MuMo's emphasis is operational: enabling decentralized publication and enforceable access control aligned with existing workflows. This differs from approaches that focus primarily on cryptographic immutability via attestation and private ledgers [18].

3.3.2. Authorization model: group-based sharing as an operational advantage

MuMo mirrors the group-based authorization model already present in the legacy dashboard: access is granted at the level of sensor groups rather than individual observations.

This is an advantage in museum practice. Loan scenarios typically require a partner to access all monitoring data relevant to a particular object/location over a period, while keeping the rest of the monitoring infrastructure private. Fine-grained authorization can be expressive, but

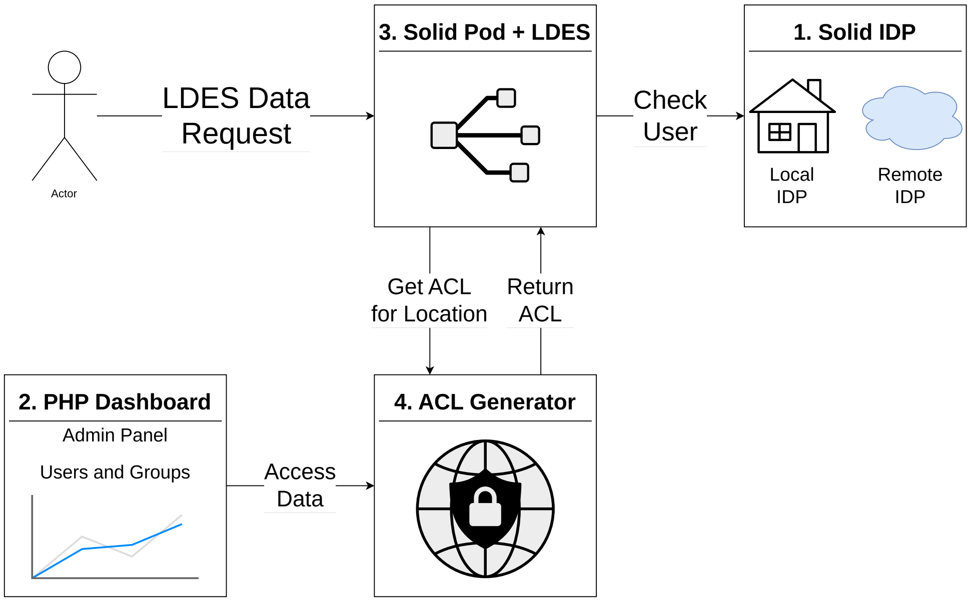

Figure 2: MuMo deployment overview with four components, an Identity Provider, a dashboard, a solid pod with the LDES publications, and an ACL file generator.

prior work highlights that it introduces substantial configuration and usability complexity in operational settings, especially for non-technical users [11].

Group-level permissions are therefore sufficient and manageable, avoiding administrative overhead that would undermine adoption.

3.3.3. Policy-aligned fragmentation (LDES + Solid together)

Solid authorization mechanisms such as Web Access Control (WAC) enforce access rules over resources and their representations [3, 4].

Because LDES allows the publisher to decide how events are partitioned into resources (fragments), MuMo chooses a fragmentation strategy that makes the group the first-class boundary: the group subtree becomes the unit of sharing.

In the deployed system, authentication is enforced at the first level of the fragment tree: users are granted access to the subtree for a given group, which implicitly includes all sensors and time slices published under that group. The corresponding ACL files are generated from the dashboard configuration; granting a user access to a group also grants access to all of that group's descendants. Because the fragment tree uses a flattened group partition, a single change in the dashboard can therefore update the effective authorization for many fragments at once.

The same structure also makes it explicit that finer-grained authorization would be technically possible (e.g., sensor-level or time-range-level restrictions), but MuMo does not activate those options to remain aligned with what museum staff can practically configure in the authoritative dashboard. In the current deploy an ACL file exists for the first layer in Figure 1, but each entry point to a subtree could have an associated ACL file.

3.4. Operationalization: components and request flow

Figure 2 shows how MuMo bridges the legacy monitoring dashboard with Solid-based identity and authorization at runtime. The key operational principle is that the dashboard remains the authoritative interface for staff, while its group-based decisions are reflected into Solid-side enforcement.

3.4.1. Runtime components

A MuMo dataspace consists of four main components.

Component 1 — Solid Identity Provider (IdP).
The IdP is where users log in (with Solid-OIDC). After login it identifies them by their WebID, which MuMo uses as the handle for access control. A deployment can run its own IdP to issue WebIDs locally, but it can also accept WebIDs from other providers, so identity does not have to be centralized.

Component 2 — PHP dashboard (admin & group management).
The dashboard is the control plane where administrators manage groups and group hierarchies (including recursive nesting), user-to-group assignments, and cross-institution sharing by granting group access to WebIDs. WebIDs may originate from the deployment's local Solid IdP or an external provider such as Inrupt; in the dashboard they function as stable identifiers for granting/revoking group access.

Component 3 — Solid Pod hosting LDES resources (data plane).
The Solid Pod hosts the published resources (including the LDES fragment tree) and enforces WAC authorization before serving protected resources. Requests are interpreted in terms of which protected subtree (group/sensor/time path) is being accessed and whether the requester's authenticated WebID is authorized.

Component 4 — ACL generator (policy materialization service).
The ACL generator bridges the dashboard's group model with Solid's resource-level authorization. It receives synchronized configuration (WebID $\rightarrow$ groups and the group hierarchy) and materializes WAC ACL documents on demand. Operationally, its function is: given a request targeting a particular group, return the effective ACL representation for that scope. This avoids manual ACL maintenance and keeps the Solid-side authorization view consistent with the dashboard configuration.

3.4.2. End-to-end request flow

1. Admin configuration (control path): an administrator updates group membership in the PHP dashboard by associating WebIDs with groups and maintaining the group hierarchy.
2. Sync to policy service: these settings are synchronized to the ACL generator as (WebID $\rightarrow$ groups) plus the group hierarchy.
3. User data request (data path): a client requests an LDES resource from the Solid Pod.
4. ACL resolution: before serving the resource, the Solid Pod determines which ACL scope applies (which group subtree is being accessed) and consults the ACL generator for the effective ACL.
5. Authentication + authorization: the Solid Pod validates identity via the relevant IdP (local or remote) and evaluates whether the authenticated WebID is granted access by the generated ACL.
6. Response: if authorized, the Solid Pod serves the requested LDES resource; otherwise it denies access.

3.4.3. Decentralized consumption and aggregation

Because each MuMo deployment publishes data independently, consumers may need to combine data from multiple sources. MuMo supports this through a client-side dashboard that retrieves sensor descriptions first, determines the user's authorized scope, and then incrementally consumes only the relevant observation streams. This avoids centralized aggregation while still enabling a unified user experience, consistent with the principle that integration occurs at the point of use.

4. In-Use Scenarios

This section illustrates how the MuMo dataspace architecture is used in practice by museum professionals to analyze, share, and contextualize environmental monitoring data.

4.1. Scenario 1: Analyzing Environmental Conditions Over Time

The primary use of the advanced dashboard is to support museum staff in assessing the long-term "health" of collection objects by analyzing environmental conditions over time. Users interact with the system through a web-based dashboard that allows them to filter and combine data based on:

- location (group),
- sensor or node,
- type of measurement (e.g., temperature, humidity),
- time constraints.

Using these filters, users can construct queries that follow an artwork throughout its lifecycle. For example, a conservator can analyze how environmental conditions evolved while an object was stored in one room, then exhibited in another, and later placed in temporary storage. Because changing the sensor configuration (or metadata) results in a new versioned semantic descriptions, the system can correctly associate observations with their deployment context at each point in time.

A key practical benefit of the Linked Data Event Streams (LDES) publication model is that **data filtering occurs before retrieval**. Rather than fetching all historical measurements and filtering client-side, the dashboard retrieves only the relevant fragments based on semantic relations and temporal constraints. For instance, when analyzing conditions in a specific year, only the corresponding fragments are accessed, avoiding unnecessary data transfer and improving responsiveness.

This is enabled by the fact that the event stream is published as a semantically linked fragment tree. For a given query, the dashboard can follow only those fragment relations that match the selected group, sensor, and time window, and prune all other subtrees. This makes selective data access a navigation problem rather than a centralized query problem, which fits cross-institutional settings where providers should not have to expose custom query endpoints. Instead of retrieving all fragments in Figure 1, the client filters on each level, thus only retrieving fragments along the path to the desired data. This keeps retrieval lightweight even as the overall monitoring history keeps growing.

4.2. Scenario 2: Cross-Institutional Access During Loans

A second, critical scenario concerns the monitoring of artworks during loans between museums. Lending institutions typically require access to environmental data from the borrowing museum to ensure that conservation conditions meet agreed standards, while borrowing institutions must retain control over their broader monitoring infrastructure.

In MuMo, this scenario is supported through **group-based access control aligned with Solid identities**. For a loan, the borrowing museum creates a dedicated group in the legacy dashboard and associates the relevant sensors with that group. Access to this group is then granted to specific Web-based identifiers belonging to the lending institution.

Because access control is enforced at the level of published data fragments, external users can authenticate using their own WebIDs and access only the data streams corresponding to the loan group. No centralized user management or data replication is required. Once the loan period ends, sensors should be taken out of the group, so new data is not shared.

This approach enables temporary, fine-grained sharing of monitoring data across institutional boundaries while remaining manageable for museum staff and compatible with existing workflows.

4.3. Scenario 3: Integrating Multiple Decentralized Data Sources

In addition to supporting analysis within a single monitoring deployment, the advanced dashboard demonstrates the ability to **combine data from multiple independent MuMo data sources**. Each MuMo deployment publishes its monitoring data and sensor descriptions independently, yet follows the same semantic representation and event-based publication model.

In practice, this allows users to access and analyze data originating from different museum setups within a single interface. For example, a user may compare environmental conditions across multiple exhibition spaces or institutions, provided they have the appropriate access rights. Because sensor metadata and observations are published as LDES, the dashboard can discover available sensors, determine authorization, and incrementally retrieve data from multiple sources without requiring centralized aggregation.

Beyond this demonstrated functionality, the same mechanisms also enable the **conceptual integration of external data sources** that are not part of the MuMo project, such as weather station measurements. Since both sensor metadata and observations are modeled using shared semantic standards, incorporating additional event streams would not require changes to the underlying architecture. While such external integrations have not yet been deployed, they directly informed the design of the system and illustrate how the dataspace approach supports extensibility and reuse.

4.4. Generalizing beyond monitoring systems

The scenarios above emphasize environmental monitoring data, but the broader takeaway is about interoperability between contextual datasets. Today, MuMo can derive rich analyses because it combines observations with time-aware context from the monitoring infrastructure itself (e.g., when a sensor was deployed where, and for which period). However, museums already maintain other structured context—most notably about artworks, locations, and movements—in collection management systems and repositories, such as Omeka-S or digital repository services such as D-RaaS [21].

If this object-centric information could be combined with MuMo's sensor and observation streams in an interoperable way, MuMo would become substantially more powerful. Queries could then be expressed directly in conservatorial terms—e.g., "show me a graph of the temperature experienced by this artwork"—because the system could follow links from an object to its associated locations and time windows, and from there build queries to retrieve to the relevant measurements.

At present, the necessary datasets often exist, but they are not interoperable: a conservator can typically assemble such answers only by manually aligning exports from the CMS/repository with monitoring data. MuMo's approach highlights that making these contextual links available in a structured, time-aware form would shift this effort from ad-hoc manual integration to repeatable, queryable reuse at the point of use, without requiring museums to replace their existing collection management infrastructure.

5. Observations and Lessons Learned

The in-use scenarios presented above highlight how specific architectural and modeling decisions shaped the practical use of MuMo in museum monitoring contexts. Rather than evaluating individual technologies in isolation, this section reflects on how these choices interacted with real-world constraints and practices.

5.1. Solid as an Institutional Boundary Mechanism

Across all scenarios, Solid played a central role in enabling data sharing without centralization. By treating Solid Pods as **institution-level data endpoints** rather than user-centric storage, MuMo aligned naturally with museum governance structures. Institutions retain control over their monitoring data while still enabling external access when required.

This was particularly evident in the loan scenario (Scenario 2), where Solid identities (WebIDs) allowed users from different institutions to authenticate without relying on a shared user database. Access could be granted and revoked by modifying group membership, supporting temporary collaborations without introducing new identity management workflows.

Lesson learned: Solid is most effective in this context when used to represent **organizational boundaries and responsibilities**, rather than as a personal data store.

5.2. Linked Data Event Streams for Scalable Analysis

The use of Linked Data Event Streams proved essential for handling the continuous and growing nature of monitoring data. In the analysis scenario (Scenario 1), LDES enabled the dashboard to retrieve only the relevant subsets of data based on semantic relations and temporal constraints, avoiding the need to fetch complete datasets and filter client-side.

Lesson learned: The combination of LDES and Solid is particularly effective because fragmentation simultaneously serves performance and governance needs. Fragment trees enable clients to prune irrelevant subtrees for a given query, while Solid access control can be enforced on the same subtree boundaries. This allows selective cross-institutional access without requiring providers to implement bespoke query endpoints or replicate data into centralized stores.

This incremental access model also facilitated multi-source analysis (Scenario 3), where data from multiple independent MuMo deployments could be consumed and combined without requiring centralized aggregation or bespoke query endpoints.

Lesson learned: Event-based publication is particularly well suited for long-running Digital Humanities data collection, where datasets evolve continuously and selective access is more important than expressive querying.

5.3. Group-Based Access Control as a Deliberate Design Choice

MuMo deliberately adopts **group-level access control**, even though finer-grained authorization is technically feasible within the chosen architecture. The use of Linked Data Event Streams combined with a hierarchical fragmentation strategy makes it possible to enforce access constraints at different levels of granularity, such as individual days or, with minor adaptations to the publication pipeline, per measurement type.

In practice, these options were not activated. The legacy dashboard that remains authoritative for configuration and access management does not provide mechanisms to express such fine-grained permissions. As a result, only those authorization concepts that are meaningful and manageable within existing museum workflows were reflected in the generated access control configurations.

This decision was particularly evident in the loan scenario, where group-level access proved sufficient to grant lending institutions visibility into relevant monitoring data without exposing unrelated information. Introducing finer-grained access control would have increased technical complexity without corresponding benefits for end users.

Lesson learned: In applied Digital Humanities settings, the appropriate granularity of access control is determined less by technical capability than by the expressive power of existing organizational tools and practices. Designing for feasible access control is often more valuable than designing for maximal access control.

5.4. Semantic Modeling as an Enabler of Integration

Semantic representations based on shared observation models enabled MuMo to separate **data production** from **data consumption**. Sensor descriptions, deployment context, and observations could be published independently, versioned over time, and interpreted consistently across systems.

This separation was critical in Scenario 1, where sensor relocations needed to be reflected correctly in longitudinal analyses, and in Scenario 3, where data from multiple sources could be combined as long as they adhered to the same semantic structures.

Lesson learned: The value of semantic modeling in practice lies less in expressiveness and more in its ability to support evolution, reuse, and alignment across independently managed systems.

5.5. Dataspace Integration at the Point of Use

Finally, the combination of Solid, LDES, semantic modeling, and client-side aggregation reflects a broader dataspace principle: **integration happens at the point of use**, not through centralized infrastructure.

Similar architectural patterns can be observed in other domain-specific data space initiatives, such as the Flanders Smart Data Space, where semantic standardization and Linked Data Event Streams are combined to support decentralized data publication across organizational boundaries [14].

More broadly, European data space initiatives emphasize federated trust, identity, and governance mechanisms to enable cross-organizational data exchange while preserving sovereignty, aligning with the boundary-oriented role Solid plays in MuMo [12, 20].

Rather than enforcing a single global schema or repository, MuMo enables institutions to publish data independently and allows consumers to integrate only what they need, when they need it. This approach proved compatible with legacy systems and institutional autonomy, both of which are common in Digital Humanities settings.

Lesson learned: Dataspace architectures are particularly well suited to DH environments where data is distributed, governance is decentralized, and collaboration is episodic.

6. Impact for the Digital Humanities Community

Many Digital Humanities projects rely on data that is collected continuously over long periods of time and embedded in local infrastructures, such as sensor data, annotations, or evolving metadata. In practice, such data often remains siloed within project-specific systems, limiting its reuse, comparability, and shareability across institutional boundaries.

Digital Humanities researchers have increasingly emphasized that the sustainability of digital resources is not only a technical issue but also an institutional and socio-technical one, shaped by funding horizons, maintenance practices, and governance constraints [22, 7].

The experience reported in this paper suggests that **dataspace-oriented approaches** are well suited to such settings. By allowing data to remain under the control of the institution that produces it, while enabling integration at the point of use, dataspaces provide a viable alternative to centralized platforms. This is particularly relevant for DH collaborations involving multiple partners with heterogeneous governance structures and long-lived legacy systems.

The use of **semantic representations as infrastructural connectors**, rather than as expressive modeling exercises, further supports interoperability in constrained environments. Lightweight semantic alignment enables independently managed datasets to be combined, extended, and reinterpreted over time, without requiring uniform tooling or deep semantic expertise from end users.

Finally, breaking down data silos enables new forms of longitudinal and comparative analysis that are difficult to achieve with isolated systems. The ability to combine data from multiple independent sources—under controlled access conditions—opens opportunities for richer contextualization and collaboration in a wide range of Digital Humanities scenarios.

Together, these observations point toward a pragmatic path for DH infrastructures that emphasizes decentralization, semantic interoperability, and governance-aware design over comprehensive but fragile integration solutions.

7. Conclusion and Future Work

This paper reported on MuMo (Museum Monitoring), a three-year applied research project that explored the use of dataspace principles for environmental monitoring in museum practice. By integrating semantic data publication, event-based access, and decentralized governance with existing monitoring infrastructure, MuMo addressed practical challenges related to long-running data collection, cross-institutional collaboration, and controlled data sharing.

A central outcome of the project is the demonstration that dataspace-oriented architectures can be deployed incrementally alongside legacy systems. Rather than requiring centralized platforms or uniform tooling, MuMo shows how interoperability and data reuse can be achieved while preserving institutional autonomy and established workflows. The project further illustrates that design choices grounded in conservatorial practice—such as group-based access control—can be both sufficient in practice and easier to sustain than more fine-grained alternatives.

Several limitations remain. Access control is currently restricted to group-level permissions, reflecting the expressive capabilities of the legacy dashboard rather than technical constraints of the dataspace architecture. Future work could explore finer-grained authorization mechanisms, such as temporal or measurement-level access, should corresponding user requirements arise. In addition, while the system supports aggregation across multiple MuMo deployments, integrating external data sources beyond the project has not yet been explored in deployed settings.

In conclusion, MuMo provides an in-use perspective on how Solid-based data governance, Linked Data Event Streams, and lightweight semantic modeling can support practical Digital Humanities workflows. The project highlights the value of aligning technical architectures with organizational realities and contributes concrete lessons for the design of interoperable, sustainable DH data infrastructures.

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